

Mathematics - Undergraduate Level Assessment 22

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

Statement 1 | The external direct product of cyclic groups is cyclic. Statement 2 | The external direct product of D_3 and D_4 is isomorphic to D_7 .

- A. True, True
- B. False, False
- C. True, False
- D. False, True

Question 2

(2 marks)

Statement 1 | If R is an integral domain, then $R[x]$ is an integral domain. Statement 2 | If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x)g(x)) = \deg f(x) + \deg g(x)$.

- A. True, True
- B. False, False
- C. True, False
- D. False, True

Question 3

(2 marks)

Let $(\mathbb{Z}_{10}, +, \cdot)$ be the ring of integers modulo 10, and let S be the subset of $\mathbb{Z}_{10}$ represented by $\{0, 2, 4, 6, 8\}$. Which of the following statements is FALSE?

- A. $(S, +, \cdot)$ is closed under addition modulo 10.

B. $(S, +, \times)$ is closed under multiplication modulo 10.

C. $(S, +, \times)$ has an identity under addition modulo 10.

D. $(S, +, \times)$ has no identity under multiplication modulo 10.

Question 4

(2 marks)

If the finite group G contains a subgroup of order seven but no element (other than the identity) is its own inverse, then the order of G could be

A. 27

B. 28

C. 35

D. 37

Question 5

(2 marks)

Find the maximum possible order for some element of $\mathbb{Z}_4 \times \mathbb{Z}_6$.

A. 4

B. 6

C. 12

D. 24

True / False

Question 1

(2 marks)

Every finitely generated torsion-free abelian group can be expressed as a direct sum of cyclic groups.

True

False

Question 2*(2 marks)*

If a and b are elements of a subgroup H , then the product $a * b^{-1}$ is also an element of H .

Question 3*(2 marks)*

$\sim$ is a symmetric relation on the set $A = \{1, 2, 3, 4\}$.

Question 4*(2 marks)*

If f is a linear transformation and $f(1, 1) = 1$ and $f(-1, 0) = 2$, then $f(3, 5) = 9$.

Question 5*(2 marks)*

If C is false, then it is possible for both A and B to be true simultaneously.

Long Form**Question 1***(10 marks)*

Find the point of intersection of the line $x - \frac{2}{3} = y + \frac{1}{4} = z - \frac{2}{12}$ and $x - y + z = 5$.

Type your answer here...

Question 2

(10 marks)

There are two distinct solutions x to the equation $18 + 5x^2 = 20x$. If each solution is rounded to the nearest integer, and then these two integers are multiplied together, what is the result?

Type your answer here...

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